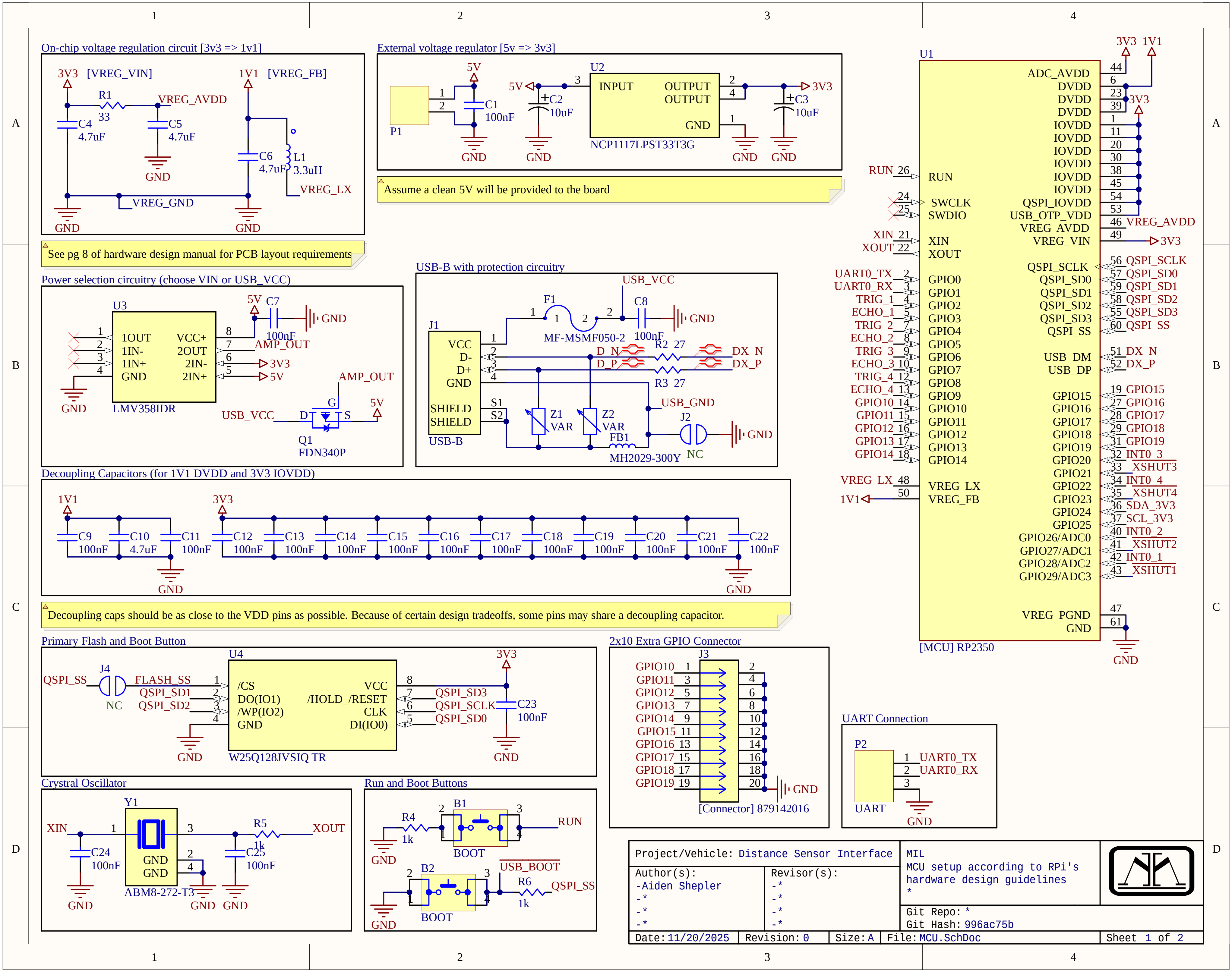


On-chip voltage regulation circuit [3v3 => 1v1]
3V3 [VREG_VIN]
1V1 [VREG_FB]
R1 33
VREG_AVDD
C4 4.7uF
C5 4.7uF
C6 4.7uF
L1 3.3uH
VREG_LX
VREG_GND
GND
See pg 8 of hardware design manual for PCB layout requirements
External voltage regulator [5v => 3v3]
P1
C1 100nF
C2 10uF
C3 10uF
U2
INPUT
OUTPUT
OUTPUT
GND
NCP1117LPST33T3G
5V
3V3
Assume a clean 5V will be provided to the board
Power selection circuitry (choose VIN or USB_VCC)
U3
1OUT
1IN-
1IN+
GND
VCC+
2OUT
2IN-
2IN+
LMV358IDR
C7 100nF
AMP_OUT
USB_VCC
Q1 FDN340P
USB-B with protection circuitry
J1
VCC
D-
D+
GND
SHIELD
SHIELD
USB-B
F1 MF-MSMF050-2
C8 100nF
D_N
D_P
R2 27
R3 27
DX_N
DX_P
Z1 VAR
Z2 VAR
FB1 MH2029-300Y
USB_GND
J2 NC
Decoupling Capacitors (for 1V1 DVDD and 3V3 IOVDD)
C9 100nF
C10 4.7uF
C11 100nF
C12 100nF
C13 100nF
C14 100nF
C15 100nF
C16 100nF
C17 100nF
C18 100nF
C19 100nF
C20 100nF
C21 100nF
C22 100nF
Decoupling caps should be as close to the VDD pins as possible. Because of certain design tradeoffs, some pins may share a decoupling capacitor.
Primary Flash and Boot Button
QSPI_SS
J4 NC
FLASH_SS
QSPI_SD1
QSPI_SD2
U4
/CS
DO(IO1)
/WP(IO2)
GND
VCC
/HOLD_/RESET
CLK
DI(IO0)
W25Q128JVSIQ TR
QSPI_SD3
QSPI_SCLK
QSPI_SD0
C23 100nF
Crystal Oscillator
XIN
Y1
ABM8-272-T3
C24 100nF
C25 100nF
R5 1k
XOUT
Run and Boot Buttons
R4 1k
B1 BOOT
RUN
B2 BOOT
USB_BOOT
R6 1k
2x10 Extra GPIO Connector
J3
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
[Connector] 879142016
UART Connection
P2
UART
UART0_TX
UART0_RX
U1
ADC_AVDD
DVDD
IOVDD
QSPI_IOVDD
USB_OTP_VDD
VREG_AVDD
VREG_VIN
RUN
SWCLK
SWDIO
XIN
XOUT
GPIO0 … GPIO25
GPIO26/ADC0
GPIO27/ADC1
GPIO28/ADC2
GPIO29/ADC3
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SS
USB_DM
USB_DP
VREG_LX
VREG_FB
VREG_PGND
GND
TRIG_1
ECHO_1
TRIG_2
ECHO_2
TRIG_3
ECHO_3
TRIG_4
ECHO_4
INT0_3
XSHUT3
INT0_4
XSHUT4
SDA_3V3
SCL_3V3
INT0_2
XSHUT2
INT0_1
XSHUT1
[MCU] RP2350
Project/Vehicle: Distance Sensor Interface
MIL
MCU setup according to RPi's hardware design guidelines
Author(s): -Aiden Shepler
Revisor(s):
Git Repo: *
Git Hash: 996ac75b
Date: 11/20/2025
Revision: 0
Size: A
File: MCU.SchDoc
Sheet 1 of 2

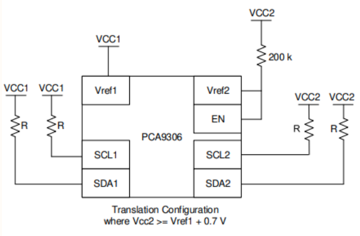
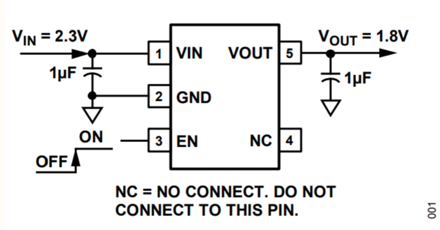
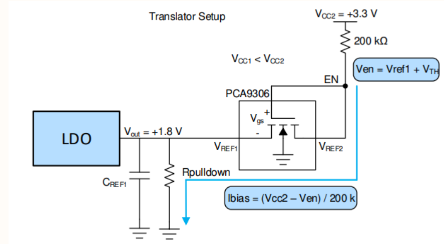
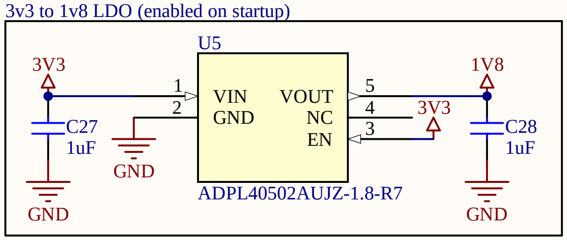


Figure 8-6. Translation Configuration

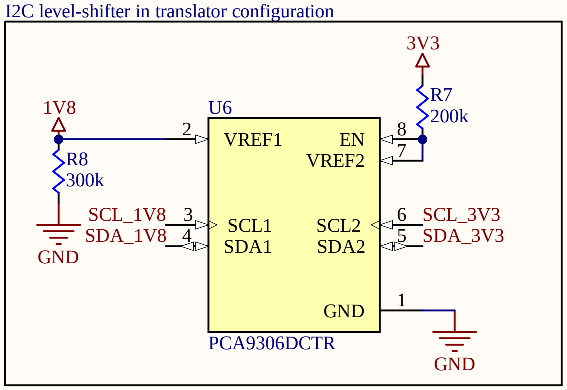




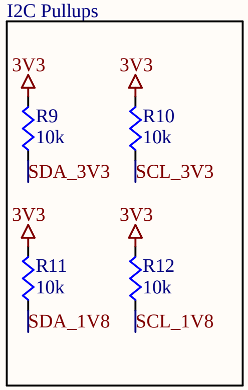
3v3 to 1v8 LDO (enabled on startup)



I2C level-shifter in translator configuration

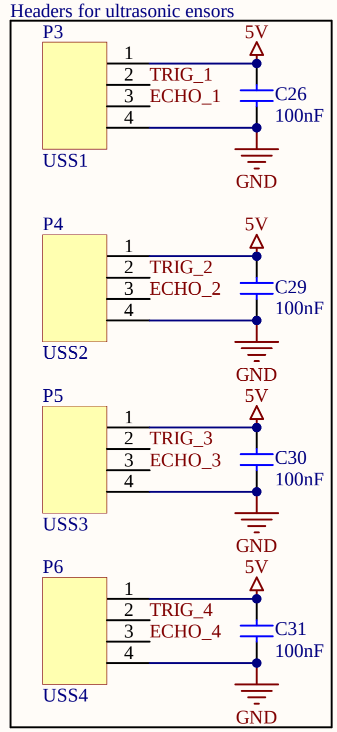


I2C Pullups

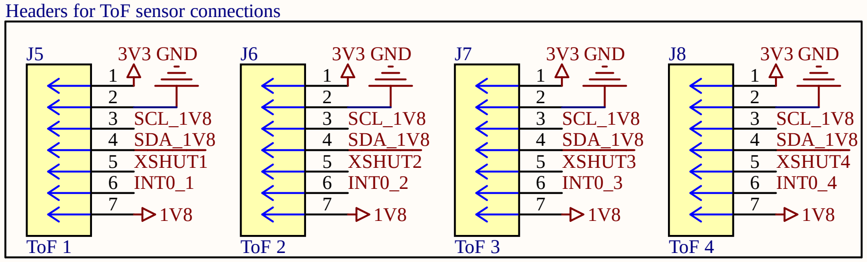


See the wiki page for the pulldown resistor calculation and why its needed.

Headers for ultrasonic ensors



Headers for ToF sensor connections



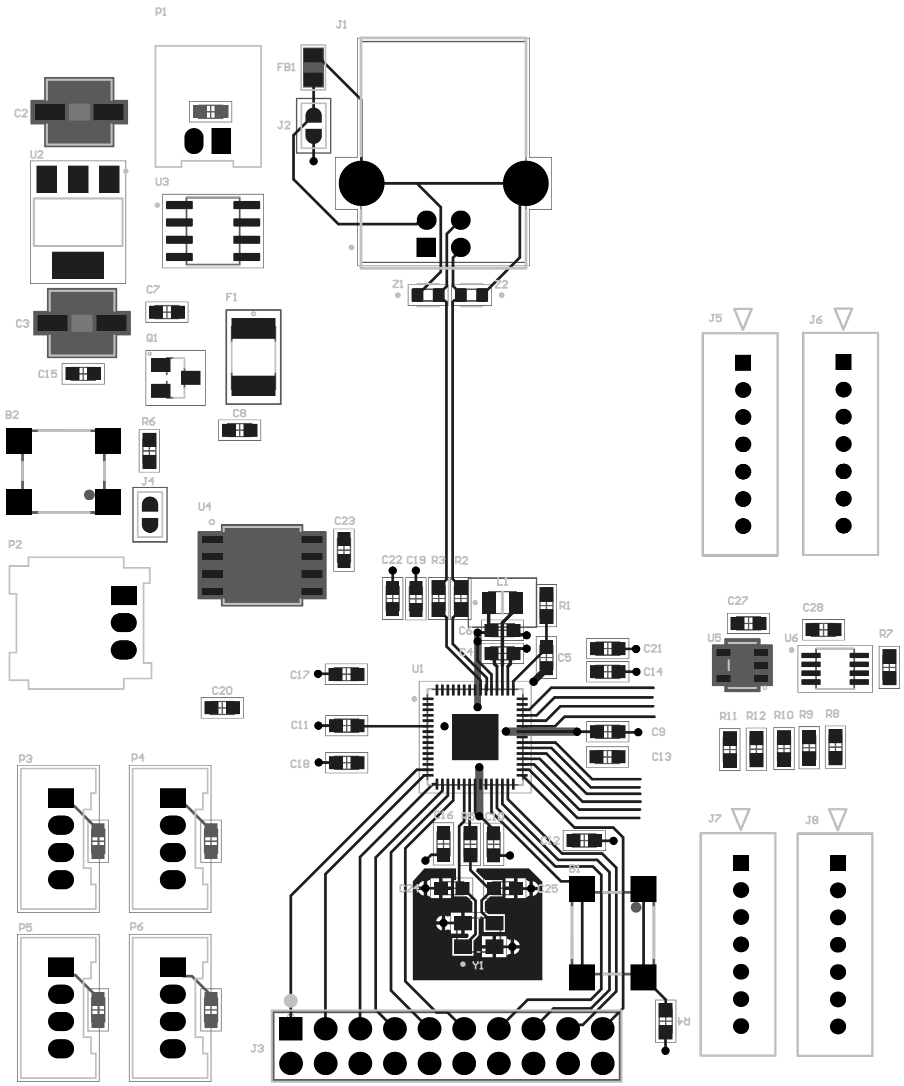
Project/Vehicle: Distance Sensor Interface

MIL
Level-shifting circuitry for ToF and headers for both ToF and ultrasonic sensors

Author(s):	Revisor(s):
-Aiden Shepler	-*
-*	-*
-*	-*
-*	-*

Git Repo: *
Git Hash: 996ac75b

Date: 11/20/2025 | Revision: 0 | Size: A | File: Distance-Sensors.SchDoc | Sheet 2 of 2


P1
J1
FB1
C2
J2
U2
U3
Z1
Z2
C7
F1
C3
J5
J6
Q1
C15
B2
R6
C8
J4
U4
C23
P2
C22
C19
R3
R2
L1
R1
C27
C28
U5
U6
R7
C21
C14
C5
C17
U1
C20
C11
C9
C13
R11
R12
R10
R9
R8
C18
P3
P4
C16
J7
J8
B1
C25
Y1
P5
P6
J3

Board Stack Report